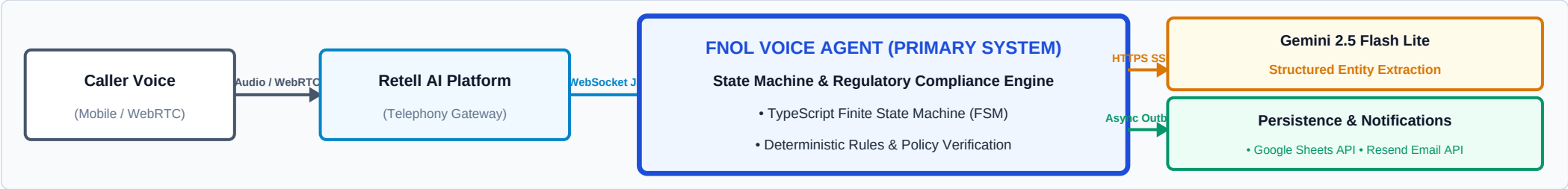


1. EXECUTIVE SUMMARY & ARCHITECTURAL PHILOSOPHY

EXECUTIVE SUMMARY: Meridian Voice AI automates motor insurance First Notice of Loss (FNOL) claims, replacing 20-minute call center queues with sub-second (<800ms) voice AI. The system pairs real-time voice streaming and single-pass LLM entity extraction with a deterministic TypeScript state machine, ensuring 100% regulatory compliance.

ARCHITECT'S PHILOSOPHY: Decouples **stochastic intelligence** (LLM entity extraction) from the **deterministic control plane** (state machine business rules). Out-of-band persistence and offloaded media processing guarantee compliance while meeting an enterprise <800ms latency budget.

2. SYSTEM ARCHITECTURE (C4 HIGH-LEVEL CONTAINER VIEW)



3. ARCHITECTURAL TRADE-OFFS

Decision	Strategic Benefit (Pros)	Current Limitation	Production Evolution
Deterministic FSM	100% compliance & zero misrouted emergencies	Rigid conversational paths	Dynamic sub-state routing
In-Memory State	Sub-ms turn latency with zero network overhead	State tied to container life	Redis Cluster + Redlock mutex
Google Sheets API	Immediate, human-readable claim portal for adjusters	Rate limits (429) & no ACID	PostgreSQL Relational DB
Flash LLM Model	Sub-350ms TTFT (~90% cheaper per turn)	Lower complex reasoning	Multi-model failover pool
Async Outbox	Unblocks voice turn audio latency (<800ms)	Risks dropped background tasks	Kafka Event Streaming

4. KEY ARCHITECTURAL DECISIONS

• Deterministic FSM over Autonomous LLM: Code owns state transitions & medical escalations to guarantee 100% regulatory compliance. • Low-Latency Gemini Extraction: Single-pass JSON extraction over SSE delivers ~350ms TTFT, staying within the <800ms budget.	• Asynchronous Persistence: Outbox pattern writes to Google Sheets and Resend out-of-band to prevent voice audio stalling. • Clear Separation of Responsibilities: Media transport, conversation state, extraction, and persistence operate as decoupled layers.
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5. FAILURE MODES & RESILIENCE MATRIX

Failure	Impact	Current Prototype Mitigation	Production Solution
Pod Restart	In-flight state reset	Cleanup on disconnect	Redis Cluster + Redlock mutex
Sheets Down	Storage delay	Local disk outbox buffer	Durable Kafka event queue
LLM Timeout	Turn stall	Secondary API failover	Circuit breaker + multi-region
Email Fail	Delayed alert	Silent log recovery	Worker retry queue with backoff

6. PRODUCTION EVOLUTION PIPELINE

PROTOTYPE (In-Memory) --> EXTERNALIZED STATE (Redis) --> DURABLE STORAGE (Postgres) --> EVENT STREAMING (Kafka) --> OBSERVABILITY (OTel) --> HORIZONTAL SCALING (K8s)